

QTL mapping of agronomically important traits in sorghum (*Sorghum bicolor* L.)

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Abstract Sorghum is one of the most important cereal crops; it is used to produce feed, sugar, and biofuel. To investigate genetic tradeoffs between grain and stem sugar production, we evaluated plant height, Brix (the percentage of soluble solids in stalk juice), 100-grain weight and flowering time over 3 years in a recombinant inbred line (RIL) population consisting of 189 individuals derived from a cross between the sweet sorghum cultivar ‘Rio’ and grain sorghum ‘BTx623’. We constructed a genetic linkage map (total length, 1418.71 cM; average distance between markers, 11.26 cM), which consisted of 118 simple sequence repeat (SSR) and 8 insertion-deletion (INDEL) markers. A total of 14 QTLs were detected on chromosomes 1, 3, 6, 7, and 9, which included 6 QTLs for plant height; 4 for Brix; and 2 QTLs for each 100-grain weight and flowering time. Eight QTLs

were detected at least in 2 years. These results will be useful for future QTL fine mapping and gene mining for these traits, and useful for the improvement of sorghum through molecular marker-assisted selection.

Keywords Sorghum · Agronomic traits · Quantitative trait loci (QTLs) · Molecular markers

Introduction

Sorghum [*Sorghum bicolor* (L.) Moench, $2n = 20$] is the fifth most widely cultivated cereal crop worldwide. With its strong drought tolerance and wide adaptability, sorghum is cultivated in arid and semi-arid regions, where it is an important food, feedstock, and biofuel energy crop (Doggett 1988). Because of its relatively small genome (about 730 Mb) and abundant genetic diversity, sorghum has become a model plant in comparative genomics of cereal crops. The availability of the complete genome sequence of the sorghum inbred line BTx623 has a substantial impact on sorghum germplasm evaluation, gene mapping and cloning, marker-assisted selection, and exploration of the mechanisms of evolution and maintenance of genetic variation in cereal crop genomes (Paterson et al. 2009).

Despite numerous studies that have mapped QTLs and generated genetic linkage maps related to such sorghum characteristics as plant height, sugar content,

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and flowering time (Mace and Jordan 2010), there are still limitations in fine mapping and cloning of genes underlying these traits.

In sorghum, four unlinked dwarfism genes, *Dw1–Dw4* (Quinby and Kapper 1954; Menz et al. 2002; Upadhyaya et al. 2013), appear to control plant height, with tall-stalk alleles partially dominant to short-stalk alleles (Lu 1999). *Dw3* encodes a P-glycoprotein auxin transporter. Genetic analysis has shown that one *Dw3* allele from the Tx430 line has an 882-bp insertion in its fifth exon (Li et al. 2015). Other genes associated with plant height, such as the genes for peroxidase 53 and an auxin transporter, have been mapped (Upadhyaya et al. 2013); however, their phenotypic effects are not clear.

Sorghum is a short-day species and the light cycle has strong effects on sorghum growth (Mace et al. 2013). Six classical main-effect loci related to flowering time, *Mal–Ma6*, have been identified (Quinby 1967; Rooney and Aydin 1999), and about 40 QTLs associated with flowering in sorghum have been reported (Mace et al. 2013). The *Mal* gene has been map-based cloned; it encodes the long-day flowering inhibitor PSEUDORESPONSE REGULATOR PROTEIN 37 (SbPRR37) (Murphy et al. 2011; Zou et al. 2012; Cuevas et al. 2016).

Brix is an important index of sorghum stalk sweetness; genomic regions affecting stalk sugar content in sorghum have been found on all chromosomes except chromosomes 8 and 10, and various QTLs have been detected depending on the mapping populations (Bian et al. 2006; Ritter et al. 2008; Murray et al. 2008; 2009; Shiringani et al. 2010; Guan et al. 2011; Rao et al. 2013; Lu et al. 2015).

The QTLs controlling grain weight have been localized to chromosomes 1–3 and 5–8 (Paterson et al. 1995; Pereira et al. 1995; Rami et al. 1998; Brown et al. 2006; Feltus et al. 2006; Srinivas et al. 2009; Mace and Jordan 2011; Rajkumar et al. 2013; Han et al. 2015; Zhang et al. 2015; Shehzad and Okuno 2015). In a previous report (Han et al. 2015), one of the QTLs, *qGW1* on chromosome 1, explained 20–40% of phenotypic variance, and was fine mapped to within 101 kb, more recently, Boyles et al. (2017) reported a novel QTL on chromosome 5, which explained > 21% of the phenotypic variance and the most significant marker was located within a putative remorin-encoding gene (Sobic.005G188400).

Here, we used a recombinant inbred line (RIL) population derived from a cross between a typical sweet sorghum line ‘Rio’ and the grain sorghum variety BTx623 to discover new loci and verify the reported results for plant height, Brix (the percentage of soluble solids in stalk juice), 100-grain weight, and flowering time. We used simple sequence repeat (SSR) and insertion-deletion (INDEL) markers to construct a genetic linkage map and identified QTLs related to these traits. The results of this study will be useful for future molecular marker-assisted breeding and QTL fine mapping.

Materials and methods

Plant materials and plot design

A RIL mapping population consisting of 189 F_{6-8} family progeny from a cross between the cultivar ‘Rio’, a high-biomass sweet sorghum, and ‘BTx623’, an elite inbred grain sorghum, was used. A field experiment was conducted at the Shenyang sorghum experimental fields of the Liaoning Academy of Agricultural Sciences, Liaoning Province, China. The two parents and the F_{6-8} RILs were planted from 2013 to 2015 at a distance of 15 cm between plants and 60 cm between rows. A total of 30 individuals in three rows were planted for each family of the RIL population with three replicates, and five individuals in the middle row were used for phenotypic measurements.

Phenotypic measurements

The plant height (in cm) was measured from the ground to the top of the panicle at maturity stage. Brix was assayed at the fourth node from the ground at the dough stage according to Zhou et al. (2005) using a PAL-1 handheld refractometer (Atago, Tokyo, Japan). Flowering time was measured as 50% anthesis for each line. After harvest, the weight of 100 filled grains (in grams) was determined.

Statistical analyses of the phenotypic data and heritability

One-way ANOVA was conducted between the parents, and broad-sense heritability [$h^2 = V_G/V_P$, where

$V_G = V_{RI} - V_E$, $V_E = 1/2(V_{P1} + V_{P2})$, $V_P = V_{RI}$] were calculated in Excel 2007. The variability between different years in the RILs were calculated by correlation.

SSR and INDEL marker analysis

To screen polymorphism between the parents, a total of 798 sorghum SSR primer pairs (Li et al. 2009; Yonemaru et al. 2009; Reddy et al. 2012; Han et al. 2015) distributed throughout the 10 chromosomes in addition, 73 INDEL primer pairs from the reduced-representation sorghum genome sequence data collected from 634 natural populations by our laboratory (our unpublished data) were used. The polymorphic loci were used to genotype the RIL population. All primers were synthesized by Invitrogen Trading Co., Ltd. (Shanghai, China).

The total genomic DNA of the parents and all RILs was extracted and purified using the improved CTAB method (Murray and Thompson 1980) from leaves collected during the heading stage. To amplify the markers, PCR was conducted in a Bio-Rad T100 Thermal cycler PCR Amplifier (Hercules, CA, USA). The PCR mixtures (25 μ L) contained DNA (1 μ L, 20 ng/ μ L), forward and reverse primers (1 μ L each, 20 pmol/ μ L), 12.5 μ L of 2 $\times$ Taq PCR Master Mix, and 9.5 μ L of ddH₂O. PCR conditions were as follows: pre-denaturation at 94 °C for 3 min followed by 34 cycles of denaturation at 94 °C for 30 s, annealing at 55–60 °C (dependent on the primers) for 30 s, extension at 72 °C for 45 s, a final extension at 72 °C for 5 min, and hold at 4 °C. The PCR products were detected by using 6% polyacrylamide gel electrophoresis and silver staining.

The PCR condition and gel electrophoresis for 73 INDEL markers were same as the SSR markers described above.

Linkage map construction and QTL analysis

The genetic linkage map construction, QTL mapping, and gene effect analysis were performed using QTL IciMapping 4.0 software (<http://www.isbreeding.net/software/?type=detail&id=18>), and a LOD score of ≥ 2.5 was used to define the existence of a QTL (Sun et al. 2013; Yin et al. 2015). The inclusive composite interval mapping (ICIM) methods (Sun

et al. 2013; Yin et al. 2015) was used to detect QTLs for each trait.

Results and discussion

Population and phenotype

There were significant differences (ANOVA; data not shown) between the parents, which resulted in rich phenotypic variation among the RIL population in plant height, Brix, 100-grain weight, and flowering time (Table 1). Although these traits were affected by the growth year, the distributions of phenotype frequencies were approximately normal for all four traits (Fig. 1). Significant positive correlation was found for each of the four traits among years, although it was smaller for Brix (Table 2). Notably, the Brix values of RILs and the parents were smaller in 2014 than in the other years, which was caused mainly by more abundant rainfall in 2014. The broad-sense heritability of the four traits ranged from 72.33 to 99.54% (Table S1); these very high values may be explained by the use of a RIL population.

Molecular linkage map construction

Among the 789 SSR primer pairs, 118 (15.0%) detected polymorphism between the parents. Of the 73 INDEL primer pairs, 8 (11.0%) showed polymorphism. These 126 polymorphic loci were used to genotype the RIL population (Table S2). A linkage map constructed based on the 118 SSR and 8 INDEL markers had a total length of 1418.71 cM, the interval between two adjacent markers ranged from 0.00 to 94.60 cM, with an average of 11.26 cM (Fig. 2).

QTL analysis

QTLs related to plant height

A total of six QTLs related to plant height were detected, with the contribution rates varying from 2.53 to 40.43%. Our QTL mapping results were different from those reported by Murray et al. (2008), although we used a RIL population derived from the same parents ('BTx623' and 'Rio'), probably because our RIL population was developed and tested in a different

Table 1 Trait values for the parental and recombinant inbred lines (RILs) of Rio × BTx623 at Shenyang, China in 3 years

Trait	2013			2014			2015		
	BTx623	Rio	RILs mean (SD) ^a	RILs range	BTx623	Rio	RILs mean (SD) ^a	RILs range	RILs mean (SD) ^a
Plant height (cm)	117	323	212 (5.7)	95–368	125	336	206 (4.6)	66–357	188 (14.1)
Brix (°brix)	14.0	21.5	14.5 (1.0)	3.1–22.8	9.8	22.1	11.3 (1.2)	1.4–21.7	15.7 (1.9)
One hundred grain weight (g)	na ^b	na ^b	na ^b	na ^b	2.10	1.77	1.90 (0.03)	0.80–3.40	2.07 (0.04)
Flowering time (days)	na ^b	na ^b	na ^b	na ^b	69	93	82(5)	67–104	81 (8)
									67–103

^aStandard deviation in parentheses^bNot assayed

location (Shengyang, China vs. Weslaco, Texas) and in different years.

The QTL qPH-1 was detected between SSR markers txp114 and txp16 on chromosome 3 with a contributions rate of 3.82–5.18% in 2014 and 2015, and the QTL qPH3-2 was detected between SSR markers txp38 and txp16 on chromosome 3 and had a contribution rate of 5.35%. The alleles from ‘Rio’ increased plant height in an additive manner, whereas the ‘BTx623’ alleles reduced plant height. The QTL qPH3-2 region included Sobic.003G379500, a gene homologous to the rice semi-dwarf gene *sd-1* (Spielmeyer et al. 2002), but no sequence differences between the coding regions of the parents were found (our unpublished data), although the promoter region still needs to be sequenced.

Two QTLs were located on chromosome 6: qPH6-1 between SSR markers sam68460 and sam27218 (33,963,730–39,669,651) and qPH6-2 between txp45 and txp265 (48,362,583–50,259,864). Their contribution rates were 2.53–3.24%, and their additive effects were positive. A sorghum gene for plant height, *Dw2*, is located at 42.2 Mb on chromosome 6 within a 100-kb interval (Morris et al. 2013; Hirsch and Oldroyd 2009; Klein et al. 2008). Recently, the *Dw2* locus was found to correspond to Sobic.006G067700 (42803037–42807134), which encodes a protein kinase homologous to Arabidopsis KIPK, a member of the AGCVIII subgroup of the AGC protein kinase family (Hilley et al. 2017). qPH6-1 and qPH6-2 may correspond to *Dw2*.

The QTL qPH7-1 was located on chromosome seven and had the largest contribution (36.04–40.43%) throughout the 3 years. It was located between SSR markers sam10667 and sam41751 (59,108,888–59,274,012), in agreement with the location of *Dw3* (Multani et al. 2003; Li et al. 2015).

The QTL qPH9-1 was located on chromosome 9 between SSR markers sam42913 and sam68465 (54,343,067–57,097,254) and had a contribution rate of 11.21–19.79%. The position of qPH9-1 was the same as that of *Dw1/SbHt9.1*, which is located at 57.2 Mb on chromosome 9 (Brown et al. 2008; Morris et al. 2013; Yamaguchi et al. 2016). *Dw1* (Sobic.009G229800) was fine mapped to an interval of ~ 33 kbp using Heterogeneous Inbred Families (HIFs) segregating only for *Dw1* (Hilley et al. 2016). A polymorphism in an exon of Sobic.009G229800 created a stop codon that truncated the encoded protein

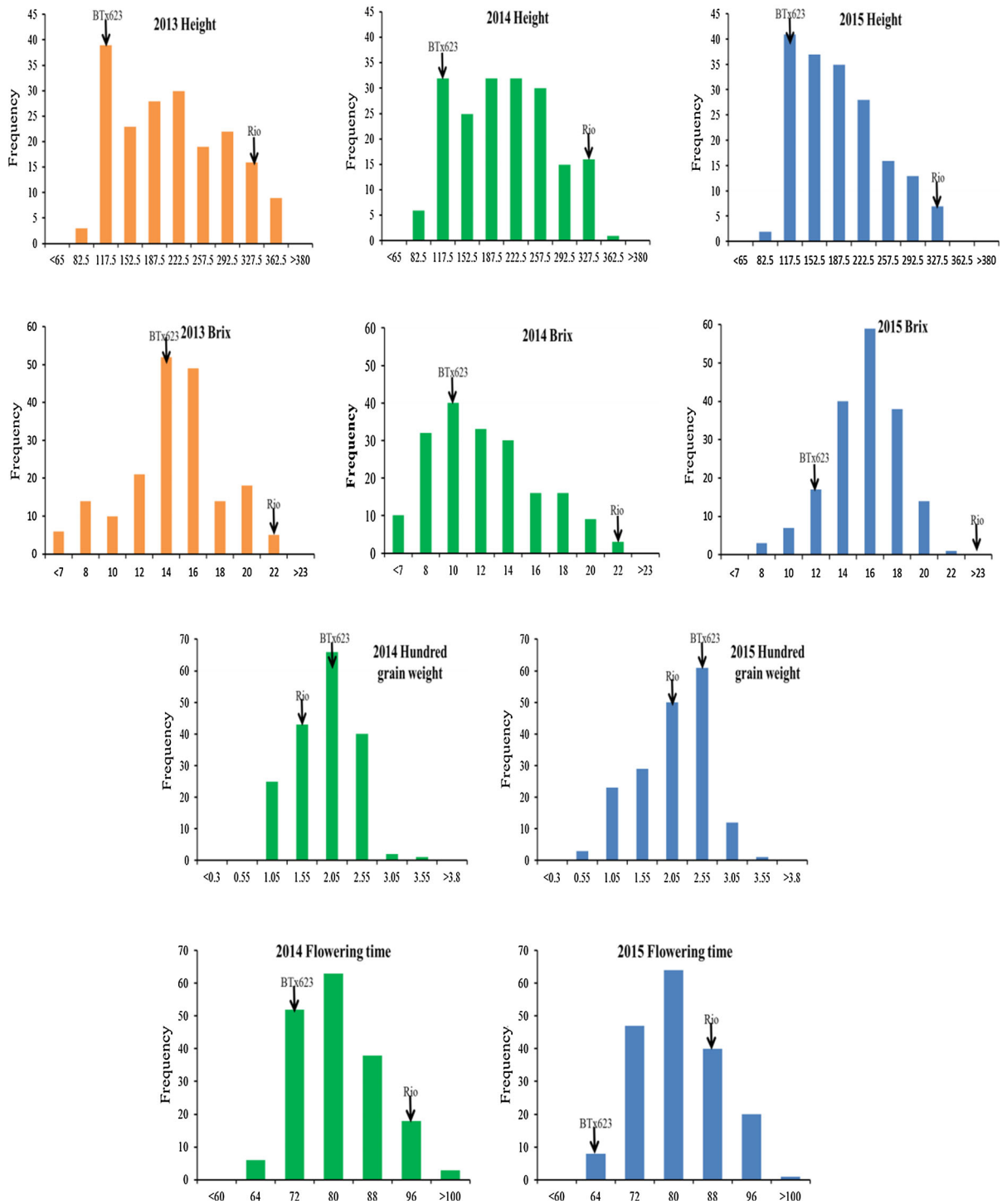


Fig. 1 Frequency distributions of plant height, Brix, 100-grain weight and flowering time in the RIL population. Arrows indicate the phenotypic values of the two parents

Table 2 Correlation coefficients between the plant height, Brix, one hundred-grain weight and flowering time in the RILs population

Trait	2013 PH	2014 PH	2015 PH	2013 B	2014 B	2015 B	2014 HGW	2015 HGW	2014 FT	2015 FT
2013 PH	1									
2014 PH	0.914**	1								
2015 PH	0.898**	0.918**	1							
2013 B	− 0.207**	− 0.178	− 0.148	1						
2014 B	0.571**	0.598**	0.582**	0.265**	1					
2015 B	0.087	0.055	0.106	0.465**	0.340**	1				
2014 HGW	0.426**	0.484**	0.434**	− 0.041	0.313**	− 0.030	1			
2015 HGW	0.470**	0.513**	0.445**	− 0.039	0.309**	− 0.099	0.904**	1		
2014 FT	0.136	0.100	0.266**	0.183	0.107	0.229	− 0.014	0.924**	1	
2015 FT	0.138	0.103	0.264**	0.187	0.101	0.226	− 0.004	− 0.052	0.899**	1

PH plant height, *b* brix, HGW one hundred grain weight, FT flowering time

*, ** Significance at the level of 0.05 and 0.01, respectively

(dw1) in sorghum cultivar 80 M (Hilley et al. 2016). It was also reported that dw1 reduces cell proliferation in internodes, and a synergistic effect of dw1 and dw3 improves lodging resistance and mechanical harvesting (Yamaguchi et al. 2016). ‘BTx623’ had three dwarf-type alleles (*dw1Dw2dw3dw4*), whereas ‘Rio’ may have the genotype *Dwdw2Dw3Dw4* (Tables 3, 4). The QTLs qPH9-1, qPH6-1, and qPH7-1 detected in this study are most probably *Dw1*, *Dw2*, and *Dw3*, respectively; whether qPH3-1 and qPH3-2 correspond to *Dw4* needs to be investigated.

QTLs related to Brix

In this study, we detected QTLs related to Brix on chromosome 1 (qB1-1; contribution, 13.06%), chromosome 6 (qB6-1, 6.90%; qB6-2, 5.85%) and chromosome 7 (qB7-1, 24.58%). All QTLs showed positive additive effects, suggesting that alleles that increased Brix were from the sweet sorghum parent ‘Rio’.

Using a RIL population derived from sweet sorghum ‘Rio’ and grain sorghum ‘BTx623’, Murray et al. (2008, 2009) detected Brix QTLs on chromosomes 1 and 3. The locations of QTLs for Brix detected in our study differed from those reported, which might reflect the complex genetic architecture underlying sugar content-related traits, which are greatly affected by the environment. Additionally, the difference in methods of sugar content determination may also have produced the discordant results.

This may explain variability in the reported locations of sugar-related QTLs and would make it problematic to use these QTLs for subsequent molecular marker-assisted breeding.

QTLs related to 100-grain weight

In this study, we detected QTLs related to 100-grain weight on chromosomes 7 (qHGW7-1; contribution 9.10–11.45%) and 9 (qHGW9-1, 18.39–20.67%). The QTLs showed a positive additive effect, indicating that the alleles that increase grain weight were derived from ‘Rio’.

Han et al. (2015) fine-mapped *qGW1* within 101 kb on chromosome 1. Boyles et al. (2017) reported a novel QTL on chromosome 5, which explained > 21% of the phenotypic variance observed in a RIL population. The most significant marker was located within a putative remorin-encoding gene (Sobic.005G188400), which is primarily expressed in early inflorescence tissues and developing embryo. But we have not detected QTLs for 100-grain weight on chromosomes 1 or 5.

The QTL intervals we detected on chromosomes 7 and 9 differed from those in the previous studies, in which QTLs for grain weight were found on all chromosomes except 5 (Paterson et al. 1995; Pereira et al. 1995; Rami et al. 1998; Brown et al. 2006; Feltus et al. 2006; Srinivas et al. 2009; Rajkumar et al. 2013). Mace and Jordan (2011) conducted a meta-analysis of 48 sorghum QTL studies published from 1995 to 2010

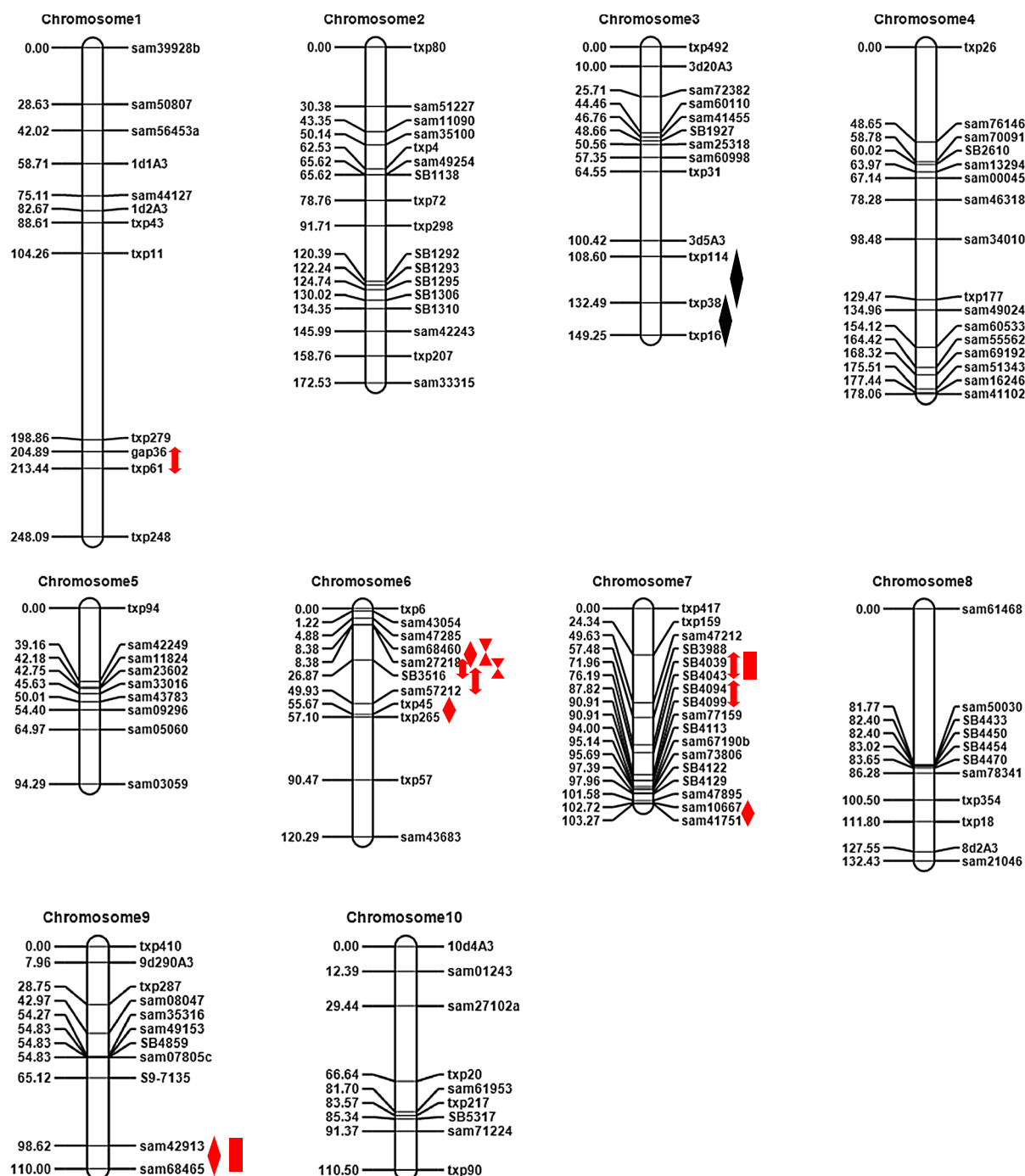


Fig. 2 Genetic map based on the 126 polymorphic SSR markers in the RIL population Note: The bars represent QTL detected by Composite interval mapping. Alleles that increased trait values inherited from BTx623 (grain parent) are shown in black. Alleles that increased trait values inherited from Rio

(sweet parent) are shown in red. Represent qPH3-1, qPH3-2, BTx623; Represent qPH6-1, qPH6-2, qPH7-1, qPH9-1, Rio; Represent qB1-1, qB6-1, qB6-2, qB7-1, Rio; Represent qHGW7-1, qHGW9-1, Rio; Represent qFT6-1, qFT6-1, Rio

Table 3 QTL of sorghum agronomically important traits detected in Rio $\times$ BTx623 RI population in three years

QTL ^a	Chr.	Flanking markers	2013				2014				2015			
			Pos. ^b	LOD ^c	PVE(%) ^d	Add ^e	Pos. ^b	LOD ^c	PVE(%) ^d	Add ^e	Pos. ^b	LOD ^c	PVE(%) ^d	Add ^e
qPH3-1	3	txp114-tpx38					132	3.82	3.82	–	130	3.53	5.18	–
qPH3-2	3	txp38-tpx16	149	6.33	5.35	–20.04								
qPH6-1	6	sam68460-sam27218	8.38	3.91	3.24	15.57					8.38	2.89	2.65	11.48
qPH6-2	6	txp45-tpx265									56	2.78	2.53	11.16
qPH7-1	7	sam10667-sam41751	103	10.15	36.04	48.03	103	10.39	37.37	49.05	103	10.13	40.43	42.92
qPH9-1	9	sam42913-sam68465	110	6.39	19.79	31.52	109	4.89	15.74	28.16	110	3.49	11.21	20.06
qB1-1	1	gap36-tpx61									212	2.51	13.06	0.91
qB6-1	6	sam27218-SB3516	26	3.52	6.90	1.22								
qB6-2	6	SB3516-sam57212					42	2.67	5.85	1.50				
qB7-1	7	SB4094-SB4099					88	5.22	24.58	2.90				
qHG7-1	7	SB4039-SB4043					76	3.24	11.45	0.17	75	2.83	9.10	0.20
qHG9-1	9	sam42913-sam68465					107	4.40	18.39	0.22	105	5.36	20.67	0.30
qFT6-1	6	sam68460-sam27218					8.38	14.18	30.58	4.73	8.38	14.80	31.49	4.74
qFT6-2	6	sam27218-SB3516					12	11.71	41.17	5.57	12	17.60	42.47	5.70

PH plant height, *B* brix, *HGW* one hundred grain weight, *FT* flowering time

^aA peak in a marker interval along the LOD profile was treated as a QTL, if there is at least one environment with the peak value higher than the LOD threshold

^bChromosomal position (cM) of the peak

^cThe number in bold was the value of a peak higher than the threshold value in the LOD profile

^dPercentage of the phenotypic variation explained by the locus at the peak positions on the LOD profile

^eAdditive effect of the identified QTL

Table 4 Summary of the identified QTL in the three environments and previous reported genes/QTL

Flanking markers	Chr.	2013 year	2014 year	2015 year	Flanking markers location	Reported or speculation genes/QTL	Gene location
txp114-tpx38	3		qPH3-1	qPH3-1	60786464–68115431		
txp38-tpx16	3	qPH3-2			68114958–70234032	Speculation Sobic.003G379500	69335724–69340213
sam68460-sam27218	6	qPH6-1		qPH6-1	33963730–39669651		
txp45-tpx265	6			qPH6-2	48362583–50259864		
sam10667-sam41751	7	qPH7-1	qPH7-1	qPH7-1	59108888–59274012	Dw3, qHT7.1	58390034, 55229509
sam42913-sam68465	9	qPH9-1	qPH9-1	qPH9-1	54343067–57097254	Dw1/SbH9.1, Sobic.009G229800	57038653–57041166
gap36-tpx61	1			qB1-1	70962186–73860271		
sam27218-SB3516	6	qB6-1			39669463–43565154		
SB3516-sam57212	6		qB6-2		43564977–46905208		
SB4094-SB4099	7		qB7-1		57197783–57424858		
SB4039-SB4043	7		qHGW7-1	qHGW7-1	38915062–46281742		
sam42913-sam68465	9		qHGW9-1	qHGW9-1	54343067–57097254		
sam68460-sam27218	6		qFT6-1	qFT6-1	33963730–39669596		
sam27218-SB3516	6		qFT6-2	qFT6-2	39669463–43565154	Mal1,SbTFL1-4, Sobic.006G068300	40305390, 42914834, 42916517

and detected nine meta-QTLs for grain weight on all chromosomes except 5 and 9. Using QTL mapping and GWAS, Zhang et al. (2015) detected four genomic regions significantly associated with grain size on chromosomes 4, 6, 7, and 10. Shehzad and Okuno (2015) located grain weight QTLs on chromosomes 1–5 and 7; overall, these QTLs explained up to 15.7% of phenotypic variation using an F₂ and F_{2:3} pedigree.

In rice, several QTLs for grain weight have been cloned: *GS3* (Fan et al. 2006), *GW2* (Song et al. 2007), *GS5* (Li et al. 2011), *GW5* (Weng et al. 2008), and *qGL3* (*OsPPKL1*, Zhang et al. 2012). But no genes homologous to the above rice genes were found in the QTL regions detected in our study. Using genome resequencing and GWAS, Tao et al. (2017) detected 114 seed-size candidate genes in sorghum, most of which were homologous to cloned seed-size genes in other crops such as rice and maize. These candidate genes reported by Tao et al. (2007) were also not located in our QTL regions.

QTLs related to flowering time

In this study, we detected two QTLs for flowering time on chromosome 6: qFT6-1 (contribution, 30.58–31.49%) and qFT6-2 (41.17–42.47%). These two QTL had a positive additive effect, indicating that the allele for extending flowering time was derived from ‘Rio’, which has a longer flowering time. The detected QTL was located between SSR markers sam27218 and SB3516 (positions 39,669,463–43,565,154). The *Mal* gene (*SbPPR37*) is located near marker txi49 at position 40,305,390 (Murphy et al. 2011; Zou et al. 2012, Cuevas et al. 2016). Our QTL is most likely the same as the *Mal* gene; its allelism to the *Mal* gene remains to be confirmed.

In conclusion, in this study, we have detected a total of 14 QTLs related to plant height, Brix, 100-grain weight and flowering time. Some QTLs have been reported previously in the same genomic regions, whereas some others are novel. The results of this study will be useful for future QTL fine-mapping and marker-assisted breeding in sorghum.

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